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UMC Reports Fourth Quarter 2021 Results

2021 EPS grew 89% YoY to NT$4.57; ROE 21.6%
Company well-positioned to capture industry growth driven by mega-trends

Fourth Quarter 2021 Overview[1]:

- **Revenue: NT$59.10 billion (US$2.14 billion)**
- **Gross margin: 39.1%; Operating margin: 29.8%**
- **Revenue from 22/28nm: 20%**
- **Capacity utilization rate: 100%+**
- **Net income attributable to shareholders of the parent: NT$15.95 billion (US$576 million)**
- **Earnings per share: NT$1.30; earnings per ADS: US$0.235**

Taipei, Taiwan, ROC – January 25, 2022 – United Microelectronics Corporation (NYSE: UMC; TWSE: 2303) ("UMC" or "The Company"), a leading global semiconductor foundry, today announced its consolidated operating results for the fourth quarter of 2021.

Fourth quarter consolidated revenue was NT$59.10 billion, increasing 5.7% QoQ from NT$55.91 billion in 3Q21. Compared to a year ago, 4Q21 revenue grew 30.5% YoY from NT$45.30 billion in 4Q20. Consolidated gross margin for 4Q21 was 39.1%. Net income attributable to the shareholders of the parent was NT$15.95 billion, with earnings per ordinary share of NT$1.30.

Jason Wang, UMC co-president, said, "In the fourth quarter, strong demand continued to drive full loading across our fabs, while overall wafer shipments grew 1.7% QoQ to 2.55 million 8-inch equivalents. For the full year, revenue in 2021 rose by more than 20% YoY and operating income reached a record high, driven by a surge in our 28nm business. The 75% YoY revenue increase from 28nm technologies strengthened our overall wafer ASP and reflected the robust chip demand related to 5G, AIoT, and automotive mega-trends. It also substantially contributed to the

[1]Unless otherwise stated, all financial figures discussed in this announcement are prepared in accordance with TIFRSs recognized by Financial Supervisory Commission in the ROC, which is different from IFRSs issued by the International Accounting Standards Board. They represent comparisons among the three-month period ending December 31, 2021, the three-month period ending September 30, 2021, and the equivalent three-month period that ended December 31, 2020. For all 4Q21 results, New Taiwan Dollar (NT$) amounts have been converted into U.S. Dollars at the December 31, 2021 exchange rate of NT$ 27.67 per U.S. Dollar.

improvement in the company's financial structure. Our healthy 28nm product pipeline will further diversify our portfolio and customer base, and enable UMC to capture additional market share."

Co-president Wang said, "Looking ahead into Q1 2022, we anticipate that demand across all nodes in UMC's addressable markets will continue to outpace supply. Our growth in the long term is supported by industry megatrends, which will be catapulted by structural changes occurring in the industry. We will continue to deepen collaboration with customers with our differentiated specialty technologies, manufacturing excellence, and capacity expansions closely linked to the demands of our partners. At the same time, we will keep pushing for cost reduction and meticulously manage our CapEx in order to deliver sustainable and healthy returns for our shareholders."

Co-president Wang added, "We were thrilled to receive a number of recognitions recently for our relentless focus on sustainability. Following our net-zero pledge in June, UMC was ranked no.1 among all semiconductor constituents listed on the Dow Jones Sustainability Indices (DJSI), which has included the company on its World Index for 14 straight years. We were also awarded Taiwan Intellectual Property Management System (TIPS) certification, Taiwan's official IP management benchmark for enterprises, underscoring our commitment to strengthen overall corporate governance. Sustainability is at the core of everything we do, and we will continue prioritizing resources to drive critical initiatives in our ESG roadmap."

Summary of Operating Results

Operating Results					
(Amount: NT$ million)	4Q21	3Q21	QoQ % change	4Q20	YoY % change
Operating Revenues	59,100	55,907	5.7	45,296	30.5
Gross Profit	23,103	20,544	12.5	10,848	113.0
Operating Expenses	(6,821)	(6,636)	2.8	(6,335)	7.7
Net Other Operating Income and Expenses	1,334	1,227	8.7	1,102	21.1
Operating Income	17,616	15,135	16.4	5,615	213.7
Net Non-Operating Income and Expenses	558	4,317	(87.1)	5,619	(90.1)
Net Income Attributable to Shareholders of the Parent	15,949	17,460	(8.7)	11,196	42.5
EPS (NT$ per share)	1.30	1.43		0.92	
(US$ per ADS)	0.235	0.258		0.166	

Fourth quarter operating revenues increased by 5.7% sequentially to NT$59.10 billion which was partly attributed to enhanced product mix and higher wafer shipments. Revenue contribution from 40nm and below technologies represented 38% of wafer revenue. Gross profit grew 12.5% QoQ to NT$23.10 billion, or 39.1% of revenue. Operating expenses was up 2.8% to NT$6.82 billion. Net other operating income reached NT$1.33 billion. Net non-operating income was NT$558 million. Net income attributable to shareholders of the parent amounted to NT$15.95 billion.

Earnings per ordinary share for the quarter was NT$1.30. Earnings per ADS was US$0.235. The basic weighted average number of outstanding shares in 4Q21 was 12,254,114,875, compared with 12,206,292,756 shares in 3Q21 and 12,206,292,756 shares in 4Q20. The diluted weighted average number of outstanding shares was 12,489,949,678 in 4Q21, compared with 12,411,100,649 shares in 3Q21 and 12,359,115,536 shares in 4Q20. The fully diluted shares counted on December 31, 2021 were approximately 12,519,315,000.

Detailed Financials Section

Operating revenues increased to NT$59.10 billion. COGS was up 1.8 % QoQ to NT$36.00 billion, which included 2.2% sequential increase in depreciation. Gross profit grew 12.5% QoQ to NT$23.10 billion, partly due to enhanced product mix. Operating expenses increased 2.8% QoQ to NT$6.82 billion, as R&D increased 3.4% QoQ to NT$3.41 billion, representing 5.8% of revenue. Net other operating income was NT$1.33 billion. In 4Q21, operating income grew 16.4% QoQ to NT$17.62 billion.

COGS & Expenses

(Amount: NT$ million)	4Q21	3Q21	QoQ % change	4Q20	YoY % change
Operating Revenues	59,100	55,907	5.7	45,296	30.5
COGS	(35,997)	(35,363)	1.8	(34,448)	4.5
Depreciation	(10,122)	(9,900)	2.2	(10,436)	(3.0)
Other Mfg. Costs	(25,875)	(25,463)	1.6	(24,012)	7.8
Gross Profit	23,103	20,544	12.5	10,848	113.0
Gross Margin (%)	39.1%	36.8%		23.9%	
Operating Expenses	(6,821)	(6,636)	2.8	(6,335)	7.7
G&A	(2,164)	(2,119)	2.2	(1,966)	10.0
Sales & Marketing	(1,240)	(1,212)	2.3	(1,175)	5.5
R&D	(3,414)	(3,303)	3.4	(3,194)	6.9
Expected Credit Impairment Gain (Loss)	(3)	(2)	37.8	0	-
Net Other Operating Income & Expenses	1,334	1,227	8.7	1,102	21.1
Operating Income	17,616	15,135	16.4	5,615	213.7

Net non-operating income in 4Q21 was NT$558 million, primarily reflecting NT$2.69 billion in net investment gain, offset by a NT$2.10 billion in other loss which included the one-time legal settlement payment and the losses of financial instrument transaction.

Non-Operating Income and Expenses

(Amount: NT$ million)	4Q21	3Q21	4Q20
Non-Operating Income and Expenses	558	4,317	5,619
Net Interest Income and Expenses	(343)	(367)	(278)
Net Investment Gain and Loss	2,689	4,534	5,703
Exchange Gain and Loss	312	164	199
Other Gain and Loss	(2,100)	(14)	(5)

In 4Q21, cash inflow from operating activities was NT$26.82 billion. Cash outflow from investing activities amounted to NT$13.34 billion, which included NT$14.95 billion in capital expenditure, resulting in free cash flow of NT$11.87 billion. Cash inflow from financing reached NT$6.37 billion, primarily from a NT$5.00 billion in bond issuance and a NT$2.90 billion payment as capacity deposit for 12A P6 offset by a NT$1.36 billion in repayment in bank loans. Net cash inflow in 4Q21 totaled NT$19.52 billion. Over the next 12 months, the company expects to repay NT$8.70 billion in bank loans.

Cash Flow Summary

(Amount: NT$ million)	For the 3-Month Period Ended Dec. 31, 2021	For the 3-Month Period Ended Sep. 30, 2021
Cash Flow from Operating Activities	26,822	22,125
Net income before tax	18,174	19,452
Depreciation & Amortization	11,817	11,754
Share of profit of associates and joint ventures	(2,491)	(3,356)
Income tax paid	(222)	(1,365)
Changes in working capital & others	(456)	(4,360)
Cash Flow from Investing Activities	(13,339)	(23,222)
Acquisition of PP&E	(14,707)	(17,457)
Acquisition of intangible assets	(100)	(371)
Decrease (Increase) in other financial assets	1,589	(5,677)
Others	(121)	283
Cash Flow from Financing Activities	6,367	(9,694)
Bank loans	(1,359)	(4,510)
Bonds Issued	5,000	11,161
Increase in deposits-in	2,896	3,738
Cash dividends	(1)	(19,870)
Others	(169)	(213)
Effect of Exchange Rate	(333)	(104)
Net Cash Flow	19,517	(10,895)
Beginning balance	113,105	124,000
Ending balance	132,622	113,105

Cash and cash equivalents increased to NT$132.62 billion. Days of inventory remained at 59 days.

Current Assets			
(Amount: NT$ billion)	4Q21	3Q21	4Q20
Cash and Cash Equivalents	132.62	113.11	94.05
Notes & Accounts Receivable	35.19	32.99	27.27
Days Sales Outstanding	53	51	55
Inventories, net	23.01	23.15	22.55
Days of Inventory	59	59	60
Total Current Assets	233.27	211.60	164.31

Current liabilities increased to NT$105.45 billion, which included a NT$39.26 billion in short-term credit/bonds. Long-term credit/bonds decreased to NT$39.83 billion. Total liabilities increased to NT$183.22 billion, leading to a debt to equity ratio of 65%.

Liabilities			
(Amount: NT$ billion)	4Q21	3Q21	4Q20
Total Current Liabilities	105.45	84.55	78.24
Notes & Accounts Payable	8.36	8.26	7.86
Short-Term Credit / Bonds	39.26	24.26	38.04
Payables on Equipment	7.88	6.72	5.45
Other	49.95	45.31	26.89
Long-Term Credit / Bonds	39.83	51.09	24.77
Long-Term Investment Liabilities	8.25	8.14	20.75
Total Liabilities	183.22	171.19	141.74
Debt to Equity	65%	66%	60%

Analysis of Revenue[2]

Revenue from Asia-Pacific increased to 66% while business from North America accounted for 21% of sales. Business from Europe was 7% while contribution from Japan remained at 6%.

Revenue Breakdown by Region

Region	4Q21	3Q21	2Q21	1Q21	4Q20
North America	21%	22%	22%	23%	29%
Asia Pacific	66%	65%	63%	63%	61%
Europe	7%	7%	8%	8%	5%
Japan	6%	6%	7%	6%	5%

Revenue contribution from 22/28nm amounted to 20% of the wafer revenue, while 40nm contribution stayed at 18% of sales.

Revenue Breakdown by Geometry

Geometry	4Q21	3Q21	2Q21	1Q21	4Q20
14nm and below	0%	0%	0%	0%	0%
14nm<x<=28nm	20%	19%	20%	20%	18%
28nm<x<=40nm	18%	18%	18%	20%	22%
40nm<x<=65nm	19%	19%	19%	18%	18%
65nm<x<=90nm	8%	8%	9%	8%	8%
90nm<x<=0.13um	12%	12%	11%	11%	11%
0.13um<x<=0.18um	13%	13%	13%	13%	13%
0.18um<x<=0.35um	7%	8%	8%	8%	8%
0.5um and above	3%	3%	2%	2%	2%

Revenue from fabless customers accounted for 86% of revenue.

Revenue Breakdown by Customer Type

Customer Type	4Q21	3Q21	2Q21	1Q21	4Q20
Fabless	86%	86%	84%	86%	87%
IDM	14%	14%	16%	14%	13%

Revenue from the communication segment remained unchanged at 46%, while business from computer applications also stayed at 17%. Business from consumer applications declined to 26% as other segments accounted represented 11% of revenue.

Revenue Breakdown by Application (1)

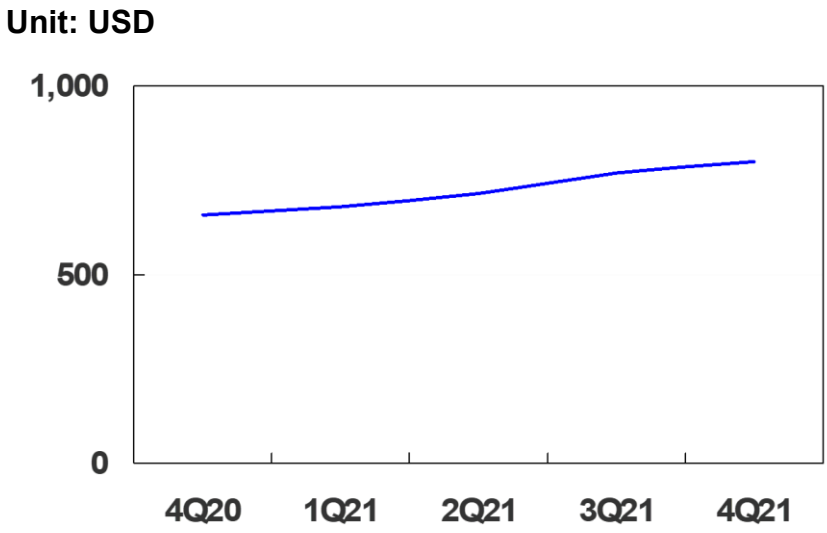
Application	4Q21	3Q21	2Q21	1Q21	4Q20
Computer	17%	17%	17%	16%	16%
Communication	46%	46%	47%	46%	49%
Consumer	26%	27%	26%	27%	25%
Others	11%	10%	10%	11%	10%

(1) ***Computer*** consists of ICs such as CPU, GPU, HDD controllers, DVD/CD-RW control ICs, PC chipset, audio codec, keyboard controller, monitor scaler, USB, I/O chipset, WLAN. ***Communication*** consists of handset components, broadband, bluetooth, Ethernet, LAN, DSP, etc. ***Consumer*** consists of ICs used for DVD players, DTV, STB, MP3/MP4, flash controller, game consoles, DSC, smart cards, toys, etc.

[2] Revenue in this section represents wafer sales

Blended ASP Trend

Blended average selling price (ASP) grew in 4Q21.



Shipment and Utilization Rate[3]

Wafer shipments increased 1.7% QoQ to 2,546K in the fourth quarter, while quarterly capacity grew to 2,419K. Overall utilization rate in 4Q21 surpassed 100%.

Wafer Shipments

	4Q21	3Q21	2Q21	1Q21	4Q20
Wafer Shipments (8" K equivalents)	2,546	2,503	2,440	2,372	2,293

Quarterly Capacity Utilization Rate

	4Q21	3Q21	2Q21	1Q21	4Q20
Utilization Rate	100%+	100%+	100%+	100%	99%
Total Capacity (8" K equivalents)	2,419	2,383	2,370	2,280	2,311

[3] Utilization Rate = Quarterly Wafer Out / Quarterly Capacity

Capacity[4]

Overall capacity in the fourth quarter grew by 1.5% QoQ to 2,419K 8-inch equivalent wafers. Capacity will remain flattish for the first quarter of 2022 at 2,420K 8-inch equivalent wafers.

Annual Capacity in *thousands of wafers*

FAB		Geometry (um)	2021	2020	2019	2018
WTK	6"	5 – 0.25	329	371	370	396
8A	8"	3 – 0.18	755	802	825	825
8C	8"	0.35 – 0.11	459	452	436	383
8D	8"	0.18 – 0.09	380	371	359	347
8E	8"	0.6 – 0.15	457	449	426	418
8F	8"	0.18 – 0.11	514	485	434	431
8S	8"	0.25 – 0.11	408	373	372	372
8N	8"	0.5 – 0.11	917	917	831	771
12A	12"	0.13 – 0.014	1,070	1,044	997	997
12i	12"	0.13 – 0.040	641	628	595	555
12X	12"	0.080 – 0.022	284	217	203	183
12M	12"	0.13 – 0.040	395	391	98	-
Total[(1)]			9,453	9,188	8,148	7,673
YoY Growth Rate			3%	13%	6%	5%

Quarterly Capacity in *thousands of wafers*

FAB	1Q22E	4Q21	3Q21	2Q21
WTK	82	81	80	84
8A	189	190	190	190
8C	113	115	115	115
8D	101	95	95	95
8E	116	115	115	115
8F	136	137	137	122
8S	109	102	102	102
8N	231	232	230	230
12A	267	271	271	271
12i	162	164	160	160
12X	77	78	74	74
12M	108	104	98	98
Total	**2,420**	**2,419**	**2,383**	**2,370**

[(1)] One 6-inch wafer is converted into 0.5625 ($6^2/8^2$) 8-inch equivalent wafer; one 12-inch wafer is converted into 2.25 ($12^2/8^2$) 8-inch equivalent wafers. Total capacity figures are expressed in 8-inch equivalent wafers.

CAPEX

CAPEX spending in 4Q21 reached US$537 million as full year 2021 CAPEX amounted to US$1.8 billion. 2022 cash-based CAPEX budget is US$3.0 billion, which including the company's collaborative Fab 12A P6 expansion plan with customers.

Capital Expenditure by Year - in US$ billion

Year	2021	2020	2019	2018	2017
CAPEX	$ 1.8	$ 1.0	$ 0.6	$ 0.7	$ 1.4

2022 CAPEX Plan

8"	12"	Total
10%	90%	US$3.0 billion

[4] Estimated capacity numbers are based on *calculated maximum output* rather than *designed capacity*. The actual capacity numbers may differ depending upon equipment delivery schedules, pace of migration to more advanced process technologies, and other factors affecting production ramp-up.

Brief Summary of Full Year 2021 Consolidated Results

- Consolidated revenue in NTD increased 20.5% YoY to NT$213.01 billion, from NT$176.82 billion in 2020.
- Gross profit increased 84.8%, compared to a year ago, representing 33.8% of 2021 revenue.
- Operating income grew 134.9% year on year, accounting for 24.3% of 2021 revenue.
- Net income attributable to stockholders of the parent increased 91.1% to NT$55.78 billion in 2021.
- EPS was NT$4.57, or EPADS of US$0.826 for 2021.
- 28nm revenue dollar grew 75% compared to 2020, as contribution from 28nm technologies and below accounted for 20% in 2021.

Operating Results			
(Amount: NT$ million)	2021	2020	YoY % change
Operating Revenues	213,011	176,821	20.5
Gross Profit	72,050	38,997	84.8
Operating Expenses	(25,590)	(23,244)	10.1
Net Other Operating Income and Expenses	5,226	6,254	(16.4)
Operating Income	51,686	22,007	134.9
Net Non-Operating Income and Expenses	10,117	5,919	70.9
Income Tax (Expense) Benefit	(6,691)	(746)	797.3
Net Income Attributable to Shareholders of the Parent	55,780	29,189	91.1
EPS (NT$ per share)	4.57	2.42	
(US$ per ADS)	0.826	0.437	

Annual Sales Breakdown in Revenue for Foundry Segment

Region	2021	2020
North America	22%	30%
Asia Pacific	65%	57%
Europe	7%	6%
Japan	6%	7%

Technology	2021	2020
14nm and below	0%	0%
14nm<x<=28nm	20%	14%
28nm<x<=40nm	18%	23%
40nm<x<=65nm	19%	17%
65nm<x<=90nm	8%	11%
90nm<x<=0.13um	12%	11%
0.13um<x<=0.18um	13%	13%
0.18um<x<=0.35um	8%	8%
0.5um and above	2%	3%

Customer Type	2021	2020
Fabless	85%	88%
IDM	15%	12%

Application	2021	2020
Computer	17%	14%
Communication	46%	52%
Consumer	27%	24%
Others	10%	10%

First Quarter 2022 Outlook & Guidance

Quarter-over-Quarter Guidance:

- Wafer Shipments: To remain flat
- ASP in USD: To increase by 5%
- Gross Profit Margin: To be approximately 40%
- Capacity Utilization: 100%
- 2022 CAPEX: US$3.0 billion

Recent Developments / Announcements

Oct. 21, 2021	UMC and suppliers join forces to build a low carbon supply chain
Nov. 13, 2021	UMC leads semiconductor foundries in 2021 Dow Jones Sustainability Indices (DJSI)
Nov. 26, 2021	Micron and UMC announce global settlement
Dec. 30, 2021	UMC awarded Taiwan Intellectual Property Management System (TIPS) Certification

Please visit UMC's website for further details regarding the above announcements

Conference Call / Webcast Announcement

Tuesday, January 25, 2022

Time: 5:00 PM (Taipei) / 5:00 AM (New York) / 10:00 AM (London)

Dial-in numbers and Access Codes:

USA Toll Free:	1-866 836-0101
Taiwan Number:	02-2192-8016
Other Areas:	+886-2-2192-8016

Access Code: UMC

A live webcast and replay of the 4Q21 results announcement will be available at www.umc.com under the “Investors / Events” section.

About UMC

UMC (NYSE: UMC, TWSE: 2303) is a leading global semiconductor foundry. The company provides high quality IC production with a focus on both logic and specialty technologies to serve every major sector of the electronics industry. UMC’s comprehensive technology and manufacturing solutions include logic/RF, embedded high voltage, embedded flash, RFSOI/BCD and IATF-16949 automotive manufacturing certification for all its manufacturing facilities. UMC operates 12 fabs that are strategically located throughout Asia with a maximum capacity of approximately 800,000 8-inch equivalent wafers per month. The company employs approximately 19,500 people worldwide, with offices in Taiwan, China, United States, Europe, Japan, Korea and Singapore. For more information, please visit: http://www.umc.com.

Safe Harbor Statements

This press release contains forward-looking statements within the meaning of Section 27A of the United States Securities Act of 1933, as amended, and Section 21E of the United States Securities Exchange Act of 1934, as amended, and as defined in the United States Private Securities Litigation Reform Act of 1995. These forward-looking statements include, but are not limited to, statements regarding anticipated financial results for the fourth quarter of 2021; the expected wafer shipment and ASP; the anticipated annual budget; capex strategies; environmental protection goals and water management strategies; impact of foreign currency exchange rates; expected foundry capacities; the ability to obtain new business opportunities; and information under the heading "Fourth Quarter of 2021 Outlook and Guidance."

These forward-looking statements involve known and unknown risks, uncertainties and other factors that may cause the actual performance, financial condition or results of operations of UMC to be materially different from what is stated or may be implied in such forward-looking statements. Investors are cautioned that actual events and results could differ materially from those statements as a result of a number of factors including, but not limited to: (i) dependence upon the frequent introduction of new services and technologies based on the latest developments in the industry in which UMC operates; (ii) the intensely competitive semiconductor, communications, consumer electronics and computer industries and markets; (iii) the risks associated with international business activities; (iv) dependence upon key personnel; (v) general economic and political conditions; (vi) possible disruptions in commercial activities caused by natural and human-induced events and disasters, including natural disasters, terrorist activity, armed conflict and highly contagious diseases; (vii) reduced end-user purchases relative to expectations and orders; and (viii) fluctuations in foreign currency exchange rates. Further information regarding these and other risk factors is included in UMC's filings with the United States Securities and Exchange Commission, including its Annual Report on Form 20-F. All information provided in this release is as of the date of this release and are based on assumptions that UMC believes to be reasonable as of this date, and UMC does not undertake any obligation to update any forward-looking statement as a result of new information, future events or otherwise, except as required under applicable law.

The financial statements included in this release are prepared and published in accordance with Taiwan International Financial Reporting Standards, or TIFRSs, recognized by the Financial Supervisory Commission in the ROC, which is different from International Financial Reporting Standards, or IFRSs, issued by the International Accounting Standards Board. Investors are cautioned that there may be significant differences between TIFRSs and IFRSs. In addition, TIFRSs and IFRSs differ in certain significant respects from generally accepted accounting principles in the ROC and generally accepted accounting principles in the United States.

- FINANCIAL TABLES TO FOLLOW -